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Reversible inhibition of gap junctional communication by tamoxifen in cultured cardiac myocytes

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Abstract Gap junction channels provide a cell-to-cell conduction pathway for direct exchange of ions and small molecules. The intercellular diffusion of a fluorescent dye, quantified in cardiac myocytes from neonatal rats by monitoring the fluorescence recovery after photobleaching, was found to be interrupted after short-term exposure (15 min) to tamoxifen, an anti-oestrogen drug often used in the treatment of human breast cancer. This diffusional uncoupling was dose dependent, occurred in the concentration range 3–25 μM and reversed after tamoxifen withdrawal. Some possible mechanisms of junctional channel closure have been examined. The cytosolic calcium concentration, examined using the fluorescent indicator Indo-1, did not vary during the short-term action of tamoxifen. A second anti-oestrogen agent (clomiphene) was able to impair gap junctional communication, whereas a third (nafoxidine) had no effect. Protein-kinase-C-inhibitor properties of tamoxifen did not seem to be involved in its uncoupling action. The characteristics of tamoxifen's action (i.e. channel inhibition delay, active concentration range, reversibility, etc.) were very similar to the previously observed effects of several other lipophilic compounds (e.g. 17 β -oestradiol, etc.) on junctional channels, and to recently reported effects of tamoxifen on voltage-gated calcium currents.

Key words Diffusional coupling · Intracellular calcium · Photobleaching · Protein kinase C · Anti-oestrogen compounds

Introduction

Gap junctions are membrane specializations that mediate intercellular communication through low-resistance cell-to-cell channels. These channels allow the direct diffusion of ions and small molecules (< 1kDa) between adja-

cent cells [9]. Found present in almost all animal tissues, gap junctional communication appears to be an important mechanism controlling cellular homeostasis, proliferation and differentiation. Cardiac, uterine and vascular smooth muscle function as electrical syncytia due to the movement of ions through junctional channels. The extent of intercellular communication can be modified by endogenous and exogenous agents, including the levels of intracellular Ca^{2+} , cAMP and pH, lipophilic agents (e.g. some alcohols, fatty acids, steroids) or transjunctional voltage, but the mechanisms of these alterations remain to be elucidated.

Gap junction channels are made up of the oligomeric assembly of protein subunits (connexins) which delineate an aqueous pore. Connexin43 (Cx43), widely distributed in different cell types, is the main gap junction protein expressed in mammalian cardiomyocytes. Different phosphorylated forms of this protein can be detected, and it has been suggested that phosphorylation of Cx43 might be involved in junctional coupling modulation (see [9]).

Tamoxifen, a synthetic nonsteroidal triphenylethylene derivative, known for its oestrogenic, anti-oestrogenic, and/or anti-tumoral actions, is widely used in the treatment of breast cancer in both pre- and post-menopausal women. The anti-tumoral character of this compound is attributed mainly to its anti-oestrogenic activity via competitive binding to the oestrogen receptor, blocking the action of oestradiol. However, several biological effects of tamoxifen cannot be reversed by oestrogen; for example, at relatively high concentrations ($\geq 1 \mu\text{M}$), tamoxifen interacts with other target molecules, such as the protein kinase C family (PKC; [7]), has a blocking effect on voltage-gated Ca^{2+} channels [10], etc. In the present study, the effects of tamoxifen on junctional communication in ventricular cardiac myocytes of neonatal rats have been examined.

Materials and methods

Cells were enzymatically dispersed from neonatal rat ventricles using trypsin, and cultured for 1–3 days, as previously described

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[2]. All measurements were made after replacement of the culture medium by a Tyrode's solution containing (in mM): NaCl 144, KCl 5.4, MgCl₂ 1, CaCl₂ 2.5, NaH₂PO₄ 0.3, glucose 5.6, HEPES 5 (buffered to pH 7.4 with NaOH) and performed at room temperature (22–24°C). Tamoxifen, clomiphene, (both the citrate salt) and nafoxidine (hydrochloride) were dissolved in dimethyl sulphoxide (DMSO) and added to the Tyrode's solution at the desired concentration. The maximal amount of DMSO in the medium (0.1% v/v) did not affect the junctional coupling of these cells in control experiments.

Quantitative measurement of cell-to-cell dye transfer

The permeability of the gap junction to the fluorescent dye 6-carboxyfluorescein (6-CF) was measured by analysing the fluorescence recovery after photobleaching (gap FRAP) of one cell in a pair or in a small group of cells as recently described [8]. Briefly, cells were exposed to a 6-CF ester, which penetrates into the cells and is hydrolysed, releasing the membrane-impermeant 6-CF. FRAP analysis was performed using a cytofluorimetric system (Anchored Cell Analysis and Sorting, ACAS 570, Meridian Instruments, Okemos, Mich., USA). In selected cells, the photobleaching of the fluorescence by strong light pulses was followed by monitoring the redistribution of unbleached dye molecules as a function of time. When the bleached cell was connected by open junctional channels to unbleached adjacent cells, its fluorescence level gradually increased, following an exponential time course (at least in the initial part of the recovery). Therefore, the rate constant k (the inverse value of the time constant) of the exponential fluorescence recovery, which provides a quantitative measure of dye transfer through junctional channels, can be obtained from the equation:

$$(F_i - F_0)/(F_i - F_0) = e^{-kt} \quad (1)$$

F_i , F_0 and F_t are the fluorescence intensities in the bleached cells, before photobleaching, immediately after and at time t after photobleaching, respectively. The rate constant k was first determined under control conditions then, using the same cell pairs or cell groups, after cultured cells had been exposed to the test solutions. Results were expressed as means $\pm$ SEM and the statistical significance of the difference of the means was established using Student's t -test for paired data, with a sufficiently large n number.

Ca²⁺ measurements

A possible involvement of a rise in the intracellular free Ca²⁺ concentration during the action of tamoxifen was examined by photometry of the light emitted by the fluorescent Ca²⁺ indicator Indo-1, as previously described [2]. Cells were loaded with the ester form of Indo-1 (Indo-1/AM, 3 μ M, 60 min) and, after careful washing to prevent further loading, the fluorescence was recorded using the cytofluorimetric system ACAS 570. Fluorescence emissions of both the free (485 nm) and the Ca²⁺-bound (405 nm) forms of the dye were collected, and their ratio (F_{405}/F_{485}) was taken as an indicator of the cytosolic Ca²⁺ concentration.

Results

Effect of tamoxifen on the cell-to-cell communication

Under control conditions, immediately after photobleaching, light emission from the selected cells was reduced to about 50% of its initial level, then a rise in fluorescence emission took place, and the rate constant k , obtained by fitting Eq. 1, was estimated at 0.24 ± 0.02 min⁻¹ ($n = 26$). In other words, the fluorescence recovery followed a monoexponential time course with a time

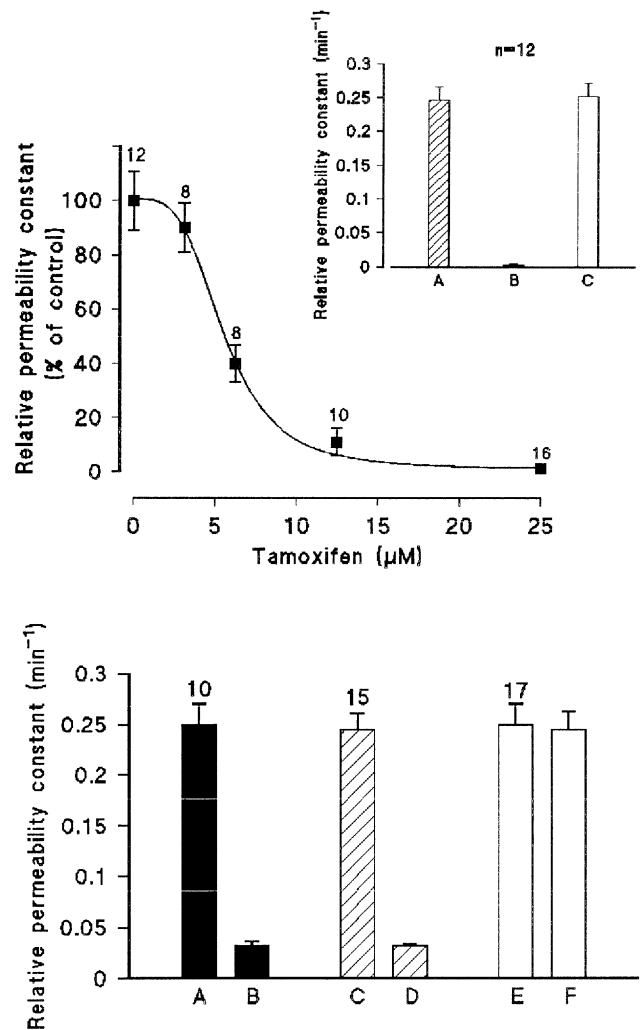


Fig. 1 Effects of anti-oestrogens on cell-to-cell dye diffusion. *Top*: dose dependence of the effect of 15-min treatments with tamoxifen, measured by means of the gapFRAP technique, in ventricular myocytes from neonatal rats. Data are expressed as a percentage ($\pm$ SEM) of the relative permeability constant (k) measured from the same cell pairs under control conditions, before tamoxifen application. *Inset*: reversibility of uncoupling after drug removal. *Bars* indicate normalized k values ($\pm$ SEM) determined under control conditions (A), after exposure to tamoxifen (25 μ M for 15 min; B) and then after washing with the control medium for 30 min (C). *Bottom*: comparison of the effects of exposures (10 μ M for 15 min) to tamoxifen (B vs A under control conditions), clomiphene (D vs C) and nafoxidine (F vs E)

constant of approximately 4 min. The rate constant determined after a 15-min exposure to tamoxifen was not affected at drug concentrations of up to 1 μ M, whereas higher tamoxifen concentrations (3–25 μ M) gradually lowered it, and a complete block of dye diffusion was observed at 25 μ M, as illustrated in Fig. 1. The half-maximal inhibitory effect of tamoxifen on intercellular communication (IC_{50}) was obtained at a concentration of approximately 5 μ M. Dye diffusion was totally restored after tamoxifen (25 μ M) removal, reflecting the reopening of the intercellular channels after wash out of the

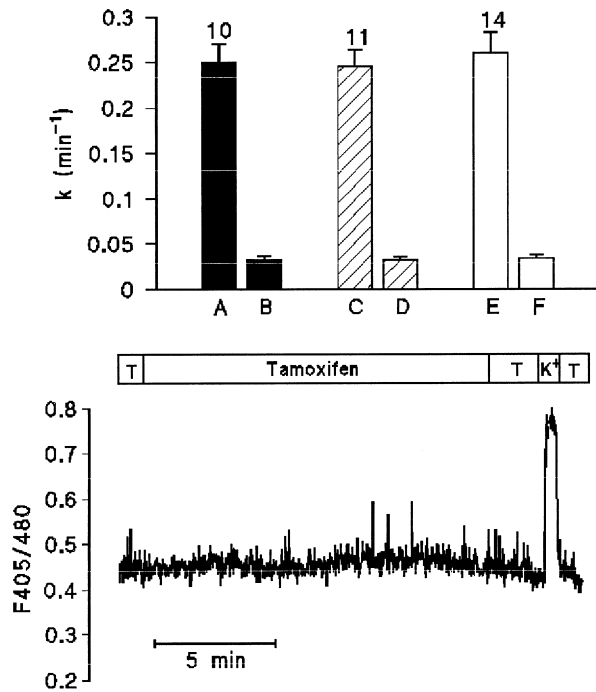


Fig. 2 Possible involvement of protein kinase C and cytosolic Ca^{2+} concentration in tamoxifen-induced junctional uncoupling. *Top*: gap junctional communication was not affected in the presence of inhibitors of protein kinase C (calphostin C; 2 μM ; C) or staurosporine (370 μM ; E) and was similarly reduced by treatments with tamoxifen (10 μM) alone (B) or co-added with calphostin C (D) or staurosporine (F). Bars indicate k values (min⁻¹), expressed as means $\pm$ SEM with (n) numbers at the top. *Bottom*: variations of intracellular Ca^{2+} concentration, measured by the fluorescence level of Indo-1 in cardiac myocytes, during the uncoupling effect of tamoxifen (25 μM) application, followed by exposure to Tyrode (T) and hyper-K⁺ (K⁺; 100 mM) solutions. After introduction of the latter medium, a rise of the intracellular Ca^{2+} level, consecutive to membrane depolarization, was recorded. Results are given as the ratio of the fluorescent emissions collected at 405 and 480 nm (F_{405}/F_{480})

drug and the absence of short-term toxicity of tamoxifen at concentrations of up to 25 μM .

Involvement of anti-oestrogenic activity in the junctional uncoupling

The effects of tamoxifen, at a concentration that causes a submaximal effect (10 μM), on cell-to-cell communication have been compared to the effects of two other oestrogen antagonists, clomiphene (Fig. 1 column D) and nafoxidine (Fig. 1 column F). Tamoxifen and chlomiphene reduced k similarly, whereas short-term exposures to nafoxidine had no effect on junctional coupling.

Involvement of protein kinases in the junctional uncoupling

Changes in the degree of cell-to-cell communication have been frequently correlated with changes in connex-

in phosphorylation (particularly, for Cx43, on serine and, to a minor extent, on threonine and tyrosine residues), which could lead to conformational changes, involving subsequent modification of the channel conductance state (for review, see [9]). As tamoxifen was able to inhibit PKC [7], the effects of tamoxifen and of two known PKC inhibitors, calphostin C and staurosporine, have been compared (Fig. 2, top). The relative permeability constants of dye transfer were not modified during short-term exposures of the cells to the PKC inhibitors calphostin C (200 nM) and staurosporine (100 nM), which could have reduced the state of connexin phosphorylation (Fig. 2, top, C and E vs A). Tamoxifen, at a concentration that causes a submaximal effect on junctional communication (10 μM), induced comparable decreases in k in the Tyrode's solution (Fig. 2 column B) and when calphostin C (Fig. 2 column D) or staurosporine (Fig. 2 column F) was present.

Ca^{2+} independence of the intercellular uncoupling

An increase in the intracellular Ca^{2+} concentration is a well-known triggering factor of gap junctional uncoupling. The possibility of involvement of Ca^{2+} in the disruption of intercellular communication by tamoxifen was examined by means of the fluorescent indicator Indo-1. The bulk cytosolic Ca^{2+} concentration did not vary during tamoxifen treatment (25 μM , Fig. 2, bottom). In contrast, a considerable rise in intracellular Ca^{2+} concentration occurred when cells were exposed to brief pulses of the high-K⁺ solution (100 mM).

Discussion

The cytosolic continuity existing between rat ventricular myocytes via gap junctional channels was rapidly and reversibly abolished when cell cultures were exposed to tamoxifen in a concentration range of 3–25 μM . This is not much above serum tamoxifen levels in patients undergoing therapy, which are in the micromolar range [7]; moreover, the concentration of this lipophilic compound in the membrane could be higher than in serum (see [10]). Junctional uncoupling was faster than the turnover of connexins in cultured cells (half-lives of 1–2 h in cardiac myocytes; [6]) and occurred without concomitant variation in the cytosolic Ca^{2+} concentration. Anti-oestrogenic properties do not seem to be involved in junctional channel closure since two compounds, tamoxifen and chlomiphene, had inhibitory effects whereas a third, nafoxidine, remained without effect under the same experimental conditions. Tamoxifen has been shown to inhibit PKC activity [7], with a relatively high IC_{50} (100 μM). PKC inhibitors (calphostin C or staurosporine) failed to alter the degree of junctional communication as well as to prevent the junctional uncoupling elicited by tamoxifen, showing that PKC does not seem to be involved in the uncoupling action of tamoxifen.

Tamoxifen also affects other membrane channels, such as L-type (IC_{50} : 2 μ M) and T-type calcium channels [10]. The characteristics of the interruption of cell-to-cell communication (uncoupling delay, active concentration range, reversibility, etc.) elicited by tamoxifen are very similar to those of 17β -oestradiol and of some other steroid compounds [5]. Both 17β -oestradiol and tamoxifen were found, in the uncoupling concentration range, to cause comparable decreases of membrane fluidity in human breast cancer cells [4]. A decrease of fluidity of the cholesterol-rich membranous domains surrounding gap junctional channels was also found to be associated with the uncoupling produced by application of heptanol [1]. These lipophilic compounds, when present at relatively high concentrations, could modify the structural organization of the membrane lipids and dramatically affect the activities of membrane proteins (enzymes, transporters, channels). These proteins are embedded in a core of lipids which remain in a gel state and maintain their tertiary structure in a particular conformation. The opening and closing of junctional channels result from conformational changes in channel proteins, and changes in the physico-chemical properties of the lipid bilayer may have profound effects on the gap junction channels and their functional state.

Tamoxifen may also elicit membrane phospholipid hydrolysis and was, for example, shown to be able, at concentrations (2.5–20 μ M) at which gap junctional communication is impaired, to induce rapid (within minutes) production of cellular phosphatidic acid (PA). In addition, a well-known junctional uncoupler, TPA (12-*O*-tetradecanoylphorbol-13-acetate), also induced PA production [3]. PA and its derivatives are important cellular second messengers governing numerous cellular events, possibly including cell-to-cell communication, by different signalling pathways, including PKC, activation of G-proteins, release of arachidonic acid, etc.

In conclusion, tamoxifen could, as could several other lipophilic compounds, interact with plasma membrane components and either interfere, by insertion into the lipid bilayer, with its arrangement and disturb the activity of membrane proteins or, alternatively, elicit membrane phospholipid hydrolysis and the resulting signal transduction cascade. A junctional uncoupling action may play a role in side-effects of tamoxifen treatments and, for example, together with the inhibitory effects on voltage-gated Ca^{2+} channels [10], may have significance for cardiovascular functions.

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References

1. Bastiaanse EML, Jongsma HJ, van der Laarse A, Takens-Kwak BR (1993) Heptanol-induced decrease in gap junctional conductance is mediated by decrease in the fluidity of membranous cholesterol-rich domains. *J Membr Biol* 136:135–145
2. Bastide B, Hervé JC, Cronier L, Déléze J (1995) Rapid onset and calcium independence of the gap junction uncoupling induced by heptanol in cultured heart cells. *Pflügers Arch* 429:386–393
3. Cabot MC, Zhang ZC, Guilano AE (1995) Tamoxifen elicits rapid transmembrane lipid signal responses in human breast cancer cells. *Breast Cancer Res Treat* 36:299–306
4. Clarke R, van der Berg HW, Nelson J, Murphy RF (1987) Pharmacological and supra pharmacological concentrations of both 17β -oestradiol and tamoxifen reduce the membrane fluidity of MCF-7 and MDA-MB-436 human breast cancer cells. *Biochem Soc Trans* 15:243–244
5. Hervé JC, Pluciennik F, Verrecchia F, Bastide B, Delage B, Joffre M, Déléze J (1996) Influence of the molecular structure of steroids on their ability to interrupt gap junctional communication. *J Membr Biol* 149:179–187
6. Laird DW, Puranam KL, Revel JP (1991) Turnover and phosphorylation dynamics of connexin43 gap junction protein in cultured cardiac myocytes. *Biochem J* 273:67–72
7. O'Brian CA, Liskamp RM, Solomon DH, Weinstein IB (1985) Inhibition of protein kinase C by tamoxifen. *Cancer Res* 45:2462–2465
8. Pluciennik F, Verrecchia F, Bastide B, Hervé JC, Joffre M, Déléze J (1996) Reversible interruption of gap junctional communication by testosterone propionate in cultured Sertoli cells and cardiac myocytes. *J Membr Biol* 149:169–177
9. Sáez JC, Berthoud VM, Moreno AP, Spray DC (1993) Gap junction. Multiplicity of controls in differentiated and undifferentiated cells and possible functional implications. In Shenolikar S, Nairn AC (eds) *Advances in second messenger and phosphoprotein research*, vol. 27. Raven, New York, pp 163–198
10. Song J, Stanley PR, Zhang F, Joshi D, Gappy S, Sowers JR, Ram JL (1996) Tamoxifen (oestrogen antagonist) inhibits voltage-gated calcium current and contractility in vascular smooth muscle from rats. *J Pharmacol Exp Ther* 277:1444–1453